

Lecture 3

The Ring R_q : Structure and Fast Arithmetic

SEC 595: Encrypted Computing

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Roadmap

Lecture 1 built $\mathbb{Z}_q$. Lecture 2 built $\mathbb{F}_p[X]$ and the NTT. This lecture puts the two together and produces the object that every scheme in the course computes in:

$$R = \mathbb{Z}[X]/(X^n + 1), \quad R_q = \mathbb{Z}_q[X]/(X^n + 1), \quad n \text{ a power of two.}$$

ML-KEM, ML-DSA, BGV, BFV and CKKS differ enormously in what they do, but they all do it here. A ciphertext is a pair of elements of R_q ; a secret key is an element of R_q with small coefficients; noise growth is a statement about $\|\cdot\|_\infty$ on R_q ; and it is fast because R_q has a CRT decomposition computable by an NTT.

The lecture has three jobs: to say precisely what “reduce modulo $X^n + 1$ ” means and why it is well defined; to prove the one inequality $\|ab\|_\infty \leq n\|a\|_\infty\|b\|_\infty$ on which every correctness proof in the FHE half of the course rests; and to build the negacyclic NTT, explaining why ML-KEM’s version of it stops one level short of ML-DSA’s.

1. Polynomials Modulo f

1.1 The Ring $\mathbb{Z}[X]/(f)$

Lecture 1 built $\mathbb{Z}_n$ by taking $\mathbb{Z}$ and doing arithmetic modulo n : add and multiply as usual, then replace the answer by its remainder on division by n . The ring $R = \mathbb{Z}[X]/(f)$ is that construction with $\mathbb{Z}[X]$ in place of $\mathbb{Z}$ and a polynomial f in place of the integer n : its elements are polynomials with integer coefficients, and its arithmetic is ordinary polynomial arithmetic followed by reduction modulo f . No ideal theory is needed here or downstream; what is used is reduction. For this to be well posed f must be *monic*, so that the division algorithm of Lecture 2, Section 1.1 works over $\mathbb{Z}$ and not merely over a field; every f we use is.

Proposition 1 (Canonical representatives). *Let $f \in \mathbb{Z}[X]$ be monic of degree n . Every $a \in \mathbb{Z}[X]$ is congruent modulo f to exactly one polynomial of degree $< n$.*

Proof. Existence: divide, $a = gf + r$ with $\deg r < n$; then $a - r$ is a multiple of f , so $a \equiv r$. Uniqueness: if r, r' both have degree $< n$ and $r - r'$ is a multiple of f , then $r - r'$ is a multiple of a degree- n polynomial and has degree $< n$, so it is 0. $\square$

So we may, and always will, identify

$$\mathbb{Z}[X]/(f) = \{a_0 + a_1X + \cdots + a_{n-1}X^{n-1} : a_i \in \mathbb{Z}\}.$$

As a set this is just $\mathbb{Z}^n$, a vector of n coefficients. What the quotient supplies on top of $\mathbb{Z}^n$ is a *multiplication*: multiply the polynomials, then reduce.

Example 1 (Reduction modulo $X^4 + 1$). Take $f = X^4 + 1$, so that $X^4 \equiv -1$, $X^5 \equiv -X$ and $X^6 \equiv -X^2$. Then

$$X^6 + 3X^5 + 2X^4 + 5X^3 + X + 4 \equiv -X^2 - 3X - 2 + 5X^3 + X + 4 = 5X^3 - X^2 - 2X + 2,$$

the coefficient vector $(2, -2, -1, 5)$ written constant-first. Reducing the coefficients modulo 17 as well gives $(2, 15, 16, 5)$ in $\mathbb{Z}_{17}[X]/(X^4 + 1)$.

The smallest interesting case is $f = X^2 + 1$, where $X^2 \equiv -1$ makes $\mathbb{Z}[X]/(X^2 + 1)$ the Gaussian integers $\mathbb{Z}[i]$, with X in the role of i : for instance $(2 + 3X)(1 + X) = 2 + 5X + 3X^2 \equiv -1 + 5X$.

1.2 The Analogy, Stated Explicitly

Every fact we prove about R_q has a companion in Lecture 1 that is already familiar, so it is worth putting the two side by side.

	Integers	Polynomials
Ambient ring	$\mathbb{Z}$	$\mathbb{Z}[X]$
Size measure	$ a $	$\deg a$
Division algorithm	$a = qn + r$, $0 \leq r < n$	$a = gf + r$, $\deg r < \deg f$
Reduce modulo	n	f , monic
Quotient	$\mathbb{Z}_n$	$\mathbb{Z}[X]/(f)$
Canonical representative	$r \in \{0, \dots, n-1\}$	r with $\deg r < n$
Inverses via	extended Euclid in $\mathbb{Z}$	extended Euclid in $\mathbb{Z}_q[X]$
Field iff	n prime	f irreducible (over a field)

The last row is the one to watch. $\mathbb{Z}_n$ is a field exactly when n is prime; $\mathbb{F}_p[X]/(f)$ is a field exactly when f is irreducible over $\mathbb{F}_p$. We shall choose q and f so that R_q is emphatically *not* a field, for reasons Section 4.3 makes precise.

One more piece of freedom, worth noticing now: the construction $\mathbb{Z}[X] \rightarrow \mathbb{Z}[X]/(X^n + 1) \rightarrow \mathbb{Z}_q[X]/(X^n + 1)$ gives the same ring in either order, since the two reductions act on independent data. L4 exploits the freedom to reduce coefficients *last* — the BFV product is computed over $\mathbb{Z}$, or a much larger modulus, and only then pushed back into R_q .

2. Why $X^n + 1$

We could reduce modulo any monic f . The whole lattice-cryptography world has settled on $f = X^n + 1$ with n a power of two. For such n this polynomial is the $2n$ -th cyclotomic polynomial Φ_{2n} , whose roots over $\mathbb{C}$ are exactly the primitive $2n$ -th roots of unity; equivalently $X^{2n} - 1 = (X^n - 1)(X^n + 1)$ splits the $2n$ -th roots of unity into the non-primitive ones (roots of $X^n - 1$) and the primitive ones (roots of $X^n + 1$). Consequently $R = \mathbb{Z}[X]/(X^n + 1) \cong \mathbb{Z}[\zeta_{2n}]$, and R is called a *power-of-two cyclotomic ring*. Four properties make it the choice, in the order they will matter to us. *Sparsity*: $X^n + 1$ has two nonzero coefficients, so reduction modulo it is free — no

multiplications, only a sign flip and an index shift (Section 3). *Norm behaviour*: its expansion factor is exactly n , the smallest possible for a degree- n ring, and Section 5 proves the bound and shows it attained. *Fast multiplication*: $X^n + 1$ splits into linear or low-degree factors over $\mathbb{Z}_q$ whenever q has enough two-adicity, giving the negacyclic NTT of Section 6 and the CRT decomposition of Section 7. *Automorphisms*: the maps $X \mapsto X^j$ for odd j are ring automorphisms of R , forming the group $\mathbb{Z}_{2n}^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_{n/2}$, and they are the rotations that implement SIMD slot permutations in L4.

3. Negacyclic Convolution

3.1 The Wrap-Around Rule

In $R = \mathbb{Z}[X]/(X^n + 1)$ the defining relation is $X^n \equiv -1$, and therefore $X^{n+r} \equiv -X^r$ for $0 \leq r < n$. Multiplication is ordinary polynomial multiplication followed by folding degrees $\geq n$ back down with a sign change.

Proposition 2 (Negacyclic convolution). *Let $a = \sum_{i < n} a_i X^i$ and $b = \sum_{j < n} b_j X^j$ in R . Then $ab = \sum_{k < n} c_k X^k$ with*

$$c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j,$$

both sums over $0 \leq i, j < n$.

Proof. The ordinary product is $\sum_{m=0}^{2n-2} d_m X^m$ with $d_m = \sum_{i+j=m} a_i b_j$. For $m < n$ the term survives; for $m = k + n$ with $0 \leq k \leq n - 2$ we have $X^m = X^n X^k \equiv -X^k$. Collecting the two contributions to X^k gives the formula. $\square$

The contrast worth remembering: modulo $X^n - 1$ the wrapped terms return with a $+$ sign (*cyclic* convolution), modulo $X^n + 1$ with a $-$ sign (*negacyclic*). One sign, and the entire NTT has to be rebuilt.

3.2 A Worked Example: $n = 4, q = 17$

We reuse the polynomials of Lecture 2, Section 6.3 and Exercise 8:

$$a(X) = 1 + 2X + 3X^2 + 4X^3, \quad b(X) = 4 + 3X + 2X^2 + X^3.$$

Step 1: the ordinary product over $\mathbb{Z}$. Convolving $(1, 2, 3, 4)$ with $(4, 3, 2, 1)$ gives $(d_0, \dots, d_6) = (4, 11, 20, 30, 20, 11, 4)$; for instance $d_3 = 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + 4 \cdot 4 = 30$.

Step 2: fold with a sign. Using $c_k = d_k - d_{k+4}$ gives $c_0 = 4 - 20 = -16$, $c_1 = 11 - 11 = 0$, $c_2 = 20 - 4 = 16$ and $c_3 = 30 - 0 = 30$, so $a \cdot b = -16 + 16X^2 + 30X^3$ in $R = \mathbb{Z}[X]/(X^4 + 1)$.

Step 3: reduce coefficients mod 17. Then $-16 \equiv 1$ and $30 \equiv 13$, giving in $R_{17} = \mathbb{Z}_{17}[X]/(X^4 + 1)$

$$a \cdot b = 1 + 16X^2 + 13X^3, \quad \text{coefficient vector } (1, 0, 16, 13).$$

In centered representatives (Lecture 1, Section 3.1) this reads $(1, 0, -1, -4)$, and $\|ab\|_\infty = 4$.

3.3 Contrast with the Cyclic Product

Lecture 2, Exercise 8 asked for $\text{NTT}^{-1}(\text{NTT}(a) \odot \text{NTT}(b))$ with $\omega = 13$ over $\mathbb{F}_{17}$, and observed that the answer is not $a \cdot b$. We can now say what it *is*. A length-4 NTT computes the product modulo $X^4 - 1$, so the wrapped terms return with a $+$ sign: $c_k = d_k + d_{k+4}$, giving $(24, 22, 24, 30)$, that is $(7, 5, 7, 13)$ modulo 17. Side by side:

$$\begin{aligned} \text{cyclic } (X^4 - 1) : & \quad (7, 5, 7, 13), \\ \text{negacyclic } (X^4 + 1) : & \quad (1, 0, 16, 13). \end{aligned}$$

They agree only in the top coefficient, which is the one that receives no wrapped contribution. Knowing both is knowing the full product: the unwrapped half is $\frac{1}{2}(\text{cyclic} + \text{negacyclic})$ and the wrapped half is $\frac{1}{2}(\text{cyclic} - \text{negacyclic})$. Exercise 2 asks you to check this.

4. The Rings R_t and R_q

4.1 Definitions

Definition 1. Fix n a power of two, a plaintext modulus $t \geq 2$ and a ciphertext modulus $q \gg t$. Set

$$R = \mathbb{Z}[X]/(X^n + 1), \quad R_t = \mathbb{Z}_t[X]/(X^n + 1), \quad R_q = \mathbb{Z}_q[X]/(X^n + 1).$$

Equivalently $R_t = R/tR$ and $R_q = R/qR$: reduce each of the n coefficients modulo t or q .

Following Lecture 1 we always take coefficients in the *centered* range, so an element of R_q is a vector of n integers in $\{-\lfloor q/2 \rfloor, \dots, \lfloor (q-1)/2 \rfloor\}$, $\|a\|_\infty$ is the largest of their absolute values, and $[a]_q$ denotes centered reduction applied coefficientwise. Two further pieces of notation: R_2 is the **ternary** distribution on coefficients $\{-1, 0, 1\}$, used for secret keys, and $\mathcal{X}$ is the error distribution, a discrete Gaussian with $\sigma = 3.2$ truncated at $\beta = 19 \approx 6\sigma$, per the Homomorphic Encryption Standard.

4.2 Units and Zero Divisors

An element of R_q is n independent coefficients, so $|R_q| = q^n$; for $q = 17$ and $n = 4$ that is $17^4 = 83521$, and for ML-KEM it is $3329^{256} \approx 2^{2995}$.

Proposition 3. *Let q be prime. Then $a \in R_q$ is a unit if and only if $\gcd(a(X), X^n + 1) = 1$ in $\mathbb{F}_q[X]$; otherwise a is a zero divisor.*

Sketch. If the gcd is 1, extended Euclid in $\mathbb{F}_q[X]$ (Lecture 2, Corollary 1) gives $ua + v(X^n + 1) = 1$, so $u = a^{-1}$ in R_q . If $g = \gcd$ is nontrivial, write $X^n + 1 = gh$ with $\deg h < n$; then $ah \equiv 0$ while $h \not\equiv 0$, so a is a zero divisor. Since R_q is finite, every element is one or the other. $\square$

Example 2 (A zero divisor in R_{17} , $n = 4$). Over $\mathbb{F}_{17}$ the polynomial $X^4 + 1$ has the four roots 2, 8, 9, 15 (Section 7.2); split them into two pairs, $a = (X - 2)(X - 8) = X^2 + 7X + 16$ and $b = (X - 9)(X - 15) = X^2 + 10X + 16$. Neither is zero, yet $ab = X^4 + 1 \equiv 0$ in R_{17} , so both are zero divisors and neither is invertible.

Example 3 (X is always a unit). $X \cdot (-X^{n-1}) = -X^n = 1$, so $X^{-1} = -X^{n-1}$ in R and in every R_q ; likewise every monomial $\pm X^j$ is a unit.

4.3 Why R_q Is Deliberately Not a Field

R_q is a field precisely when q is prime and $X^n + 1$ is irreducible over $\mathbb{F}_q$, and for n a power of two that never happens once $n \geq 4$. The degree of each irreducible factor of $X^n + 1$ over $\mathbb{F}_q$ is the multiplicative order of q in $\mathbb{Z}_{2n}^\times$; for $2n \geq 8$ that group is $\mathbb{Z}_2 \times \mathbb{Z}_{n/2}$, whose exponent is only $n/2$, so no factor reaches degree n and there are always at least two of them.

The question is therefore never *whether* R_q decomposes but *how far*, and the standards push it as far as they can. Three things depend on that decomposability. *Speed*: a complete or near-complete splitting of $X^n + 1$ is what makes the NTT possible, and irreducibility would force $O(n^2)$ multiplication. *Batching*: the CRT factors of R_t are the SIMD plaintext slots, and an irreducible $X^n + 1$ over $\mathbb{F}_t$ would give exactly one. *Composite moduli*: FHE uses $q = \prod q_i$, a product of machine-word primes, so $\mathbb{Z}_q \cong \prod \mathbb{Z}_{q_i}$ (RNS) — which fills R_q with zero divisors by construction, and that is what allows the modulus chain to be truncated one prime at a time. Lecture 1 warned that fields are not the goal; this is the payoff. The one place their absence is felt is inversion, and schemes are designed to avoid needing it.

5. Norms and the Expansion Factor

5.1 The Infinity Norm on R

Definition 2. For $a = \sum_{i < n} a_i X^i \in R$, set $\|a\|_\infty = \max_i |a_i|$. For $a \in R_q$, apply this to the centered representative: $\|a\|_\infty = \|[a]_q\|_\infty$.

This is the coefficient-wise ℓ_∞ norm on $\mathbb{Z}^n$, transplanted to R by the identification of Section 1.1. It is not multiplicative, and quantifying its failure to be multiplicative is the content of the next theorem.

5.2 The Two Inequalities

Theorem 1. For all $a, b \in R = \mathbb{Z}[X]/(X^n + 1)$:

$$(i) \quad \|a + b\|_\infty \leq \|a\|_\infty + \|b\|_\infty, \quad (ii) \quad \|a \cdot b\|_\infty \leq n \|a\|_\infty \|b\|_\infty.$$

Proof. (i) Addition is coefficientwise, so $|a_k + b_k| \leq |a_k| + |b_k|$; take the maximum over k .

(ii) By Section 3.1 the k -th coefficient of ab is $c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j$. The two sums are disjoint, and together they range over the pairs (i, j) with $i + j \equiv k \pmod{n}$. For each of the n values of i there is exactly one such j , so c_k is a signed sum of exactly n products $a_i b_j$. Hence $|c_k| \leq \sum_i |a_i| |b_{j(i)}| \leq n \|a\|_\infty \|b\|_\infty$, and taking the maximum over k gives the claim. $\square$

Definition 3. The **expansion factor** of a ring R is $\delta_R = \sup\{ \|ab\|_\infty / (\|a\|_\infty \|b\|_\infty) \}$ over nonzero a, b . Theorem 1 says $\delta_R \leq n$ for $R = \mathbb{Z}[X]/(X^n + 1)$.

The bound is not merely an upper bound: it is attained, so $\delta_R = n$ exactly. The proof of (ii) says how to attain it — make all n signed terms in some c_k agree in sign and have absolute value $\|a\|_\infty \|b\|_\infty$.

Example 4 (The bound is tight, $n = 4$). Take $a = b = 1 + X + X^2 + X^3$, so $\|a\|_\infty = \|b\|_\infty = 1$. The ordinary square has coefficients $(1, 2, 3, 4, 3, 2, 1)$, and folding with $c_k = d_k - d_{k+4}$ gives

$$ab = -2 + 0 \cdot X + 2X^2 + 4X^3, \quad \|ab\|_\infty = 4 = n \|a\|_\infty \|b\|_\infty.$$

The top coefficient $c_3 = 4$ is the one receiving no wrapped, hence no negated, contribution: all four products $a_i b_{3-i}$ are $+1$. The *cyclic* square of the same pair, $c_k = d_k + d_{k+4}$, is $4 + 4X + 4X^2 + 4X^3$, where every coefficient hits the bound.

$\delta_R = n$ is nonetheless a worst case, and a pessimistic one: for random a, b the n terms in c_k have random signs, so $|c_k|$ is around $\sqrt{n} \|a\|_\infty \|b\|_\infty$. Libraries use heuristic $\sqrt{n}$ -style bounds for parameter selection and reserve $\delta_R = n$ for provable statements; every bound below is of the provable kind.

5.3 The Payoff: Growth and Correctness

The reason to care is that noise grows, and $\delta_R = n$ is what makes it grow fast. A BFV homomorphic multiplication of ciphertexts with noises v_1, v_2 produces noise of roughly $2tn^2 \|SK\|_\infty \cdot \max(\|v_1\|_\infty, \|v_2\|_\infty)$, so each multiplication *multiplies* the noise; the n^2 is there because the operation runs two ring products in sequence, each paying a factor $\delta_R = n$. Addition, by contrast, costs only $\|v_1\|_\infty + \|v_2\|_\infty$ — inequality (i), with no factor of n at all. That asymmetry is why multiplicative depth, not circuit size, is the resource FHE parameters are budgeted against. Against this stands the correctness condition, which in BGV at level ℓ reads $\|v\|_\infty < q_\ell/(2t)$: every parameter-selection procedure in the course is the same calculation, bounding $\|v\|_\infty$ after the intended circuit using (i) and (ii) and then choosing q large enough that the inequality still holds at the last level.

Forward reference. The first place the inequality earns its keep is the fresh-ciphertext noise of BFV, where $ct = (\Delta m + PK_0 u + e_1, PK_1 u + e_2)$ evaluates at the secret to $\Delta m + (-eu + e_1 + e_2 SK)$ and two applications of (ii) bound the bracket. That derivation, in both its generic $2n\beta^2 + \beta$ and ternary $2n\beta + \beta$ forms, with the numbers for $n = 4096$, is Section 7 of the BFV lecture; it is not repeated here.

6. Roots of Unity and the Negacyclic NTT

6.1 The Problem

Lecture 2 gave us a length- n NTT over $\mathbb{F}_q$ whenever $n \mid q - 1$, and showed that transforming, multiplying pointwise, and transforming back computes a product. Section 3.3 showed which product: the *cyclic* one, modulo $X^n - 1$. That is the wrong ring, and the reason is structural. The NTT evaluates at $\omega^0, \dots, \omega^{n-1}$, the n -th roots of unity, which are the roots of $X^n - 1$; evaluation at a root of f is precisely the quotient map onto $\mathbb{F}_q[X]/(f)$. To get $X^n + 1$ we must evaluate at *its* roots instead — the primitive $2n$ -th roots of unity.

6.2 The Twist

Suppose $\mathbb{F}_q$ contains an element ψ of order exactly $2n$. Then $\omega := \psi^2$ has order n , and the roots of $X^n + 1$ in $\mathbb{F}_q$ are the odd powers of ψ , that is $\psi\omega^j$ for $j = 0, \dots, n-1$: each satisfies $(\psi\omega^j)^n = \psi^n = -1$, since ψ^n has order 2. So the negacyclic transform we want is

$$\tilde{a}_j = a(\psi\omega^j) = \sum_{i=0}^{n-1} a_i \psi^i \omega^{ij} = \sum_{i=0}^{n-1} (\psi^i a_i) \omega^{ij}.$$

Read the right-hand side: it is the *ordinary* length- n NTT of Lecture 2 applied to the twisted vector $(\psi^i a_i)_i$. No new transform is needed.

Step	Operation
1. Twist	$a_i \mapsto \psi^i a_i$, and $b_i \mapsto \psi^i b_i$
2. Transform	NTT of length n with $\omega = \psi^2$
3. Pointwise	$\tilde{c}_j = \tilde{a}_j \tilde{b}_j$
4. Inverse transform	NTT $^{-1}$ of length n with ω^{-1} and n^{-1}
5. Untwist	$c_i \mapsto \psi^{-i} c_i$

Proposition 4. *The five steps above compute $a \cdot b$ in $\mathbb{Z}_q[X]/(X^n + 1)$.*

Sketch. Steps 1–2 compute $a(\psi\omega^j)$ for all j , i.e. the image of a under the ring map $\mathbb{F}_q[X]/(X^n + 1) \rightarrow \prod_j \mathbb{F}_q[X]/(X - \psi\omega^j)$. That map is a ring isomorphism (Section 7.1), so pointwise multiplication of the images is the image of the product, and steps 4–5 invert it. $\square$

The extra cost over a cyclic transform is $2n$ scalar multiplications for the twist and n for the untwist, with precomputed powers of ψ ; in practice the twist folds into the butterflies of the first NTT layer and costs nothing. Appendix A carries out all five steps for $n = 4$, $q = 17$.

6.3 Why $2n \mid q - 1$, Not $n \mid q - 1$

The construction needs ψ of order $2n$. By Lecture 1, Section 5.1, $\mathbb{F}_q^\times$ is cyclic of order $q - 1$ and contains an element of order d if and only if $d \mid q - 1$, so the requirement is

$$2n \mid q - 1.$$

An element of order n is not enough: ω alone evaluates at the roots of $X^n - 1$, and the roots of $X^n + 1$ are simply not in the group it generates. Since n is a power of two, $2n \mid q - 1$ says the two-adicity of q (Lecture 2, Section 5) is at least $\log_2 n + 1$.

Example 5 ($n = 4$, $q = 17$). Here $q - 1 = 16$ and $2n = 8 \mid 16$. Lecture 2 recorded that 3 is a primitive root of $\mathbb{F}_{17}$, so $\psi = 3^2 = 9$ has order 8: indeed $9^2 = 81 \equiv 13$, $9^4 \equiv 13^2 = 169 \equiv 16 \equiv -1$ and $9^8 \equiv 1$. Then $\omega = \psi^2 = 13$ has order 4 — the same ω as in Lecture 2’s worked NTT over $\mathbb{F}_{17}$, so the transform of Section 6.2 is Lecture 2’s with a twist bolted on. The four elements of order 8 are the odd powers 9, 15, 8, 2 of ψ , and any serves as ψ .

7. Complete versus Incomplete Splitting

Section 6 built the NTT assuming $X^n + 1$ falls apart into n linear factors. That is not automatic: how far it falls apart is decided by q alone, and it decides how many layers the transform runs and what its “pointwise” step multiplies.

Theorem 2 (CRT for R_q). *Let q be prime and suppose $X^n + 1 = f_1 f_2 \cdots f_r$ with the f_i pairwise coprime in $\mathbb{F}_q[X]$. Then*

$$\mathbb{Z}_q[X]/(X^n + 1) \cong \prod_{i=1}^r \mathbb{F}_q[X]/(f_i),$$

by $a \mapsto (a \bmod f_1, \dots, a \bmod f_r)$, respecting $+$ and $\times$ componentwise.

Proposition 5. *Let q be a prime with $q \equiv 1 \pmod{4}$, let e be its two-adicity (so $2^e \mid q - 1$ and $2^{e+1} \nmid q - 1$), and let $n = 2^k$. Then $X^n + 1$ factors over $\mathbb{F}_q$ into*

$$r = \min(n, 2^{e-1})$$

irreducible factors, each of degree n/r and each of the form $X^{n/r} - r_i$.

The NTT of Section 6 is the case $r = n$: every factor linear, every quotient a copy of $\mathbb{F}_q$, and “reduce modulo $X - r_i$ ” is just “evaluate at r_i ”, so the transform *is* the CRT map. When $r < n$ the transform runs $\log_2 r$ layers instead of $\log_2 n$ and stops early, leaving a pointwise step that multiplies polynomials of degree n/r modulo $X^{n/r} - r_i$.

Example 6 (Complete splitting: $n = 4$ over $\mathbb{F}_{17}$). Here $q - 1 = 16 = 2^4$, so $e = 4$ and Proposition 5 predicts $r = \min(4, 2^3) = 4$ linear factors.

The roots of $X^4 + 1$ are the elements of order 8, which are the odd powers of $\psi = 9$ from Section 6:

$$\psi = 9, \quad \psi^3 = 15, \quad \psi^5 = 8, \quad \psi^7 = 2.$$

Each satisfies $x^4 = -1$: for instance $9^2 = 81 \equiv 13$ and $13^2 = 169 \equiv 16 \equiv -1$. So

$$X^4 + 1 = (X - 2)(X - 8)(X - 9)(X - 15) \quad \text{over } \mathbb{F}_{17}.$$

Check by coefficients: the roots sum to $2 + 8 + 9 + 15 = 34 \equiv 0$, matching the absent X^3 term, and their product is $2 \cdot 8 \cdot 9 \cdot 15 = 2160 = 127 \cdot 17 + 1 \equiv 1$, matching the constant term. ✓

The CRT map sends a to its four residues mod $X - r_i$, and reducing a polynomial modulo $X - r_i$ leaves the constant $a(r_i)$. For $a = 1 + 2X + 3X^2 + 4X^3$ these four numbers are

$$(a(9), a(15), a(8), a(2)) = (16, 11, 13, 15),$$

which is exactly $\text{NTT}(\tilde{a})$ from Appendix A. *The NTT and the CRT decomposition are the same map.* Multiplication in each component is multiplication of scalars in $\mathbb{F}_{17}$ — one integer multiply per slot, which is why the complete case is the fast one.

Example 7 (Incomplete splitting: $n = 4$ over $\mathbb{F}_{13}$). Here $q - 1 = 12 = 2^2 \cdot 3$, so $e = 2$ and $8 \nmid 12$: there is no element of order 8, hence $X^4 + 1$ has no root in $\mathbb{F}_{13}$ at all. Proposition 5 predicts $r = \min(4, 2^1) = 2$ factors of degree 2.

Since $5^2 = 25 \equiv -1 \pmod{13}$ we may write $X^4 + 1 \equiv X^4 - 25$ and use the difference of squares:

$$X^4 + 1 = (X^2 - 5)(X^2 + 5) \quad \text{over } \mathbb{F}_{13}.$$

Expanding gives $X^4 - 25 \equiv X^4 + 1$, as $-25 \equiv 1 \pmod{13}$. ✓ Neither quadratic splits further: that would need a square root of 5 or of $8 \equiv -5$, and the squares modulo 13 are $\{1, 3, 4, 9, 10, 12\}$, containing neither.

Multiplying through the two components. Take the running pair $a = 1 + 2X + 3X^2 + 4X^3$ and $b = 4 + 3X + 2X^2 + X^3$, now over $\mathbb{F}_{13}$. Reduce each modulo $X^2 - 5$, where $X^2 \equiv 5$ and $X^3 \equiv 5X$:

$$a \equiv (1 + 3 \cdot 5) + (2 + 4 \cdot 5)X = 16 + 22X \equiv 3 + 9X, \quad b \equiv (4 + 2 \cdot 5) + (3 + 1 \cdot 5)X = 14 + 8X \equiv 1 + 8X.$$

Multiply the two linear polynomials, again using $X^2 \equiv 5$:

$$(3 + 9X)(1 + 8X) = 3 + 33X + 72X^2 \equiv (3 + 72 \cdot 5) + 33X = 363 + 33X \equiv 12 + 7X \pmod{13}.$$

Modulo $X^2 + 5$, where $X^2 \equiv -5$, the same steps give $a \equiv 12 + 8X$, $b \equiv 7 + 11X$, and

$$(12 + 8X)(7 + 11X) = 84 + 188X + 88X^2 \equiv (84 - 88 \cdot 5) + 188X = -356 + 188X \equiv 8 + 6X \pmod{13}.$$

Cross-check. The direct negacyclic product of a and b over $\mathbb{F}_{13}$ is $(10, 0, 3, 4)$, i.e. $10 + 3X^2 + 4X^3$. Reducing it modulo $X^2 - 5$ gives $(10 + 3 \cdot 5) + (0 + 4 \cdot 5)X = 25 + 20X \equiv 12 + 7X$, and modulo $X^2 + 5$ gives $(10 - 15) + (0 - 20)X \equiv 8 + 6X$. Both components agree. ✓

Each pointwise step here cost four integer multiplications instead of one. That is the whole price of incomplete splitting — and it is the routine FIPS 203 calls `BaseCaseMultiply`.

7.1 The Same Contrast at Full Size

Exactly this distinction, at $n = 256$, separates the two NIST standards.

	ML-DSA	ML-KEM
Modulus q	8380417	3329
$q - 1$	$2^{13} \cdot 1023$	$2^8 \cdot 13$
Two-adicity e	13	8
Factors r	$\min(256, 2^{12}) = 256$	$\min(256, 2^7) = 128$
Degree of each	1 (linear)	2 (quadratic)
NTT layers	8	7
Pointwise step	scalar product	<code>BaseCaseMultiply</code>

Forward reference. The same theorem, applied to R_t rather than R_q , is *batching*. If $X^n + 1$ splits over $\mathbb{F}_t$ into r factors of degree d , then $R_t \cong \prod_{i=1}^r \mathbb{F}_{t^d}$ and a single plaintext carries r independent slots that add and multiply in parallel. L4, Section 5 builds this; the CKKS analogue over $\mathbb{C}$ gives $n/2$ complex slots, because conjugate roots are paired.

8. Modules over R_q

ML-KEM and ML-DSA do not work with single ring elements but with short vectors of them. This section supplies that one piece of notation — and nothing about hardness, which is L5.

Definition 4. R_q^k is the set of k -tuples $(s_1, \dots, s_k)$ with $s_i \in R_q$, added and scaled componentwise. It is a free **module** of rank k over R_q : the definition of a vector space with the field replaced by a ring.

A matrix $A \in R_q^{k \times k}$ acts by $(As)_i = \sum_j A_{ij} s_j$, each $A_{ij} s_j$ a negacyclic product, so As costs k^2 ring multiplications. Norms extend the obvious way, $\|s\|_\infty = \max_i \|s_i\|_\infty$, and

$$\|As\|_\infty \leq k n \|A\|_\infty \|s\|_\infty$$

by Theorem 1 applied to each of the k terms of each entry. The module's expansion factor is kn , which is exactly the dimension of the corresponding lattice.

Definition 5 (Module-LWE, informally). Sample $A \leftarrow R_q^{k \times k}$ uniformly and s, e small from a distribution on R_q^k . An **MLWE sample** is the pair (A, t) with $t = As + e$. The decision problem is to distinguish it from (A, u) for uniform $u \in R_q^k$.

Two special cases are already familiar: $n = 1$ makes the ring $\mathbb{Z}_q$ and this is plain LWE in dimension k ; $k = 1$ makes it Ring-LWE in R_q . MLWE interpolates, and the underlying lattice has dimension nk .

Example 8 (A toy MLWE sample over R_{17} , $n = 4$, $k = 2$). Every object is a 2-vector or 2×2 matrix of elements of $R_{17} = \mathbb{Z}_{17}[X]/(X^4 + 1)$, each itself 4 coefficients; so A holds $k^2 n = 16$ integers

and s, e, t hold $kn = 8$ each. Writing polynomials as coefficient vectors, constant first, take

$$A = \begin{pmatrix} (3, 5, 11, 2) & (9, 1, 4, 14) \\ (7, 12, 2, 6) & (15, 8, 10, 1) \end{pmatrix}, \quad s = \begin{pmatrix} (1, -1, 0, 1) \\ (-1, 0, 1, -1) \end{pmatrix}, \quad e = \begin{pmatrix} (0, 1, -1, 0) \\ (1, 0, 0, -1) \end{pmatrix},$$

with s and e ternary, so $\|s\|_\infty = \|e\|_\infty = 1$.

One product in full. Compute $A_{11}s_1$, the negacyclic product of $(3, 5, 11, 2)$ with $(1, -1, 0, 1)$. Using $c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+4} a_i b_j$:

$$\begin{aligned} c_0 &= 3 \cdot 1 - (5 \cdot 1 + 11 \cdot 0 + 2 \cdot (-1)) = 3 - 3 = 0, \\ c_1 &= (3 \cdot (-1) + 5 \cdot 1) - (11 \cdot 1 + 2 \cdot 0) = 2 - 11 = -9 \equiv 8, \\ c_2 &= (3 \cdot 0 + 5 \cdot (-1) + 11 \cdot 1) - 2 \cdot 1 = 6 - 2 = 4, \\ c_3 &= 3 \cdot 1 + 5 \cdot 0 + 11 \cdot (-1) + 2 \cdot 1 = -6 \equiv 11, \end{aligned}$$

so $A_{11}s_1 = (0, 8, 4, 11)$. The other three products, computed the same way, are

$$A_{12}s_2 = (5, 6, 2, 12), \quad A_{21}s_1 = (1, 3, 1, 11), \quad A_{22}s_2 = (0, 1, 6, 9).$$

Assemble. Add within each row, then add e (reduced mod 17, so $(0, 1, 16, 0)$ and $(1, 0, 0, 16)$):

$$As = ((5, 14, 6, 6), (1, 4, 7, 3)), \quad t = As + e = ((5, 15, 5, 6), (2, 4, 7, 2)) \in R_{17}^2.$$

The pair (A, t) is the sample and s is the secret. Note that t shows no trace of s being ternary — that is the whole point — and that the underlying lattice has dimension $nk = 8$.

8.1 Why ML-KEM Varies k and Not n

ML-KEM ships three parameter sets differing only in k : ML-KEM-512, -768 and -1024 take $k = 2, 3, 4$ at $n = 256$, for lattice dimensions $nk = 512, 768, 1024$. Three reasons to hold n fixed. *One NTT*: changing n changes the ring, hence the required two-adicity of q , hence q , the reduction constants and the whole transform, whereas changing k changes a loop bound — so one arithmetic core, and one certification, covers all three levels. *Granularity*: with power-of-two n the only way to raise security through n is to double it, whereas raising k moves in steps of n . *Less structure*: a rank- k module lattice has fewer subfields and automorphisms to exploit than a degree- nk ring, so MLWE hedges against Ring-LWE's extra structure helping an attacker — while FHE takes the opposite bet, $k = 1$ with n in the thousands, because it needs that structure for batching and rotations.

Forward reference. Hardness has been deferred entirely. Why $As + e$ should be indistinguishable from uniform, how MLWE reduces to a worst-case lattice problem, and how the approximation factor depends on q/β — that is L5. This section supplies only the algebra those statements are written in.

Summary

1. $\mathbb{Z}[X]/(f)$ for monic f of degree n is polynomials with arithmetic done modulo f : multiply, then reduce. Each class has one representative of degree $< n$, so the ring is $\mathbb{Z}^n$ with a twisted multiplication — Lecture 1's $\mathbb{Z}_n$ with $\mathbb{Z}[X]$ in place of $\mathbb{Z}$.

2. For n a power of two, $X^n + 1 = \Phi_{2n}(X)$, whose roots are the primitive $2n$ -th roots of unity. It is chosen for sparsity, small expansion factor, NTT-friendliness, and its automorphism group.
3. In $\mathbb{Z}[X]/(X^n + 1)$, $X^n = -1$, so multiplication is negacyclic convolution: $c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j$. For $a = 1 + 2X + 3X^2 + 4X^3$, $b = 4 + 3X + 2X^2 + X^3$ over $\mathbb{Z}_{17}$ this gives $(1, 0, 16, 13)$, against $(7, 5, 7, 13)$ cyclically.
4. R_t and R_q are R with coefficients reduced, taken centered; $|R_q| = q^n$, so $|R_{17}| = 83521$ at $n = 4$. An element is a unit iff it is coprime to $X^n + 1$; R_q is deliberately not a field, and its zero divisors give the NTT, batching, and the RNS modulus chain.
5. $\|a+b\|_\infty \leq \|a\|_\infty + \|b\|_\infty$ and $\|ab\|_\infty \leq n\|a\|_\infty \|b\|_\infty$, the latter because each product coefficient is a signed sum of exactly n terms. The expansion factor $\delta_R = n$ is attained, by $a = b = 1 + X + X^2 + X^3$ at $n = 4$. Hence the BFV multiplication growth factor $\approx 2tn^2\|SK\|_\infty$ and the BGV correctness condition $\|v\|_\infty < q_\ell/(2t)$.
6. Negacyclic multiplication by NTT: twist by ψ^i with ψ of order $2n$, run the length- n transform with $\omega = \psi^2$, multiply pointwise, invert, untwist. The requirement is $2n \mid q-1$, since ω alone reaches only the roots of $X^n - 1$. Appendix A works the case $n = 4$, $q = 17$, $\psi = 9$, $\omega = 13$.
7. For $q \equiv 1 \pmod{4}$ with two-adicity e , CRT decomposes R_q into $r = \min(n, 2^{e-1})$ factors of degree n/r . Over $\mathbb{F}_{17}$, $X^4 + 1 = (X-2)(X-8)(X-9)(X-15)$; over $\mathbb{F}_{13}$ it has no root and splits only as $(X^2 - 5)(X^2 + 5)$. ML-DSA gets 256 linear factors, ML-KEM 128 quadratics.
8. R_q^k is a free module of rank k ; an MLWE sample is $(A, As + e)$ with $A \in R_q^{k \times k}$, and its lattice has dimension nk . ML-KEM varies $k \in \{2, 3, 4\}$ at fixed $n = 256$ so that one NTT core serves all three security levels.

Exercises

1. Work in $R = \mathbb{Z}[X]/(X^4 + 1)$. Reduce $X^7 + 3X^5 - X^4 + 2$ to its canonical representative of degree < 4 , and write down explicitly the multiple of $X^4 + 1$ by which the two differ. Then explain why the same procedure would fail for $f = 2X^4 + 1$.
2. With a, b as in Section 3, negacyclic product $c^- = (1, 0, 16, 13)$ and cyclic product $c^+ = (7, 5, 7, 13)$ over $\mathbb{Z}_{17}$: verify that $2^{-1}(c^+ + c^-)$ recovers the low half $(d_0, \dots, d_3)$ of the ordinary product mod 17, and $2^{-1}(c^+ - c^-)$ the high half $(d_4, d_5, d_6, 0)$. Why is this a special case of CRT?
3. Verify that $\{2, 8, 9, 15\}$ are exactly the elements of order 8 in $\mathbb{F}_{17}^\times$. Redo Appendix A's five steps with $\psi = 2$ instead of $\psi = 9$ (so $\omega = 4$) and confirm you again obtain $(1, 0, 16, 13)$. Which intermediate vectors change, and why is the answer forced?
4. Compute ab for $a = b = 1 + X + X^2 + X^3$ in $\mathbb{Z}[X]/(X^4 + 1)$ and confirm $\|ab\|_\infty = 4$, so that the bound of Theorem 1(ii) is tight. Then explain why no choice of a, b with $\|a\|_\infty = \|b\|_\infty = 1$ can exceed it, and compute the cyclic product of the same pair for comparison.
5. An implementation uses $n = 4096$, $\beta = 19$ and $t = 1024$. Evaluate the generic fresh-noise bound $2n\beta^2 + \beta$ and the ternary bound $2n\beta + \beta$. Using $\|v\|_\infty < q/(2t)$, find the smallest integer $\log_2 q$ admitting a *fresh* ciphertext under each, and say how many bits the loose bound costs.

6. Factor $X^4 + 1$ over $\mathbb{F}_5$. First use Proposition 5 to predict the number and degree of the factors, then find them explicitly and verify by expanding. (Hint: which elements of $\mathbb{F}_5$ square to -1 ?)
7. Falcon uses $q = 12289$ with $n = 512$. Compute the two-adicity of q and decide whether $X^{512} + 1$ splits completely over $\mathbb{F}_{12289}$. How many NTT layers does Falcon run, and how does its pointwise step compare with ML-KEM's?
8. Prove that $X^{-1} = -X^{n-1}$ in R_q for every q and verify it for $n = 4$. Then, from the factorisation of $X^4 + 1$ over $\mathbb{F}_{17}$ in Section 7.2, exhibit a zero-divisor pair in R_{17} other than the one in Section 4.2.
9. For ML-KEM-768 ($n = 256$, $k = 3$, $q = 3329$), count the $\mathbb{Z}_q$ elements needed to write down A , s and e , and the coefficient multiplications in computing As naively. Compare with the plain-LWE instance of the same lattice dimension $nk = 768$, where A is a 768×768 matrix over $\mathbb{Z}_q$: what is the ratio of key sizes, and of multiplication counts?
10. Let $q \equiv 1 \pmod{2n}$ so that $R_q \cong \mathbb{F}_q^n$. Show that R_q has $(q - 1)^n$ units, and deduce that the fraction of non-invertible elements is $1 - (1 - 1/q)^n \approx n/q$ for $n \ll q$. Evaluate this for ML-DSA's parameters, and say what it implies for a scheme that must invert a uniformly random ring element.